Vaxcel®

CEFOBACTAM 1g
INJECTION

MAL11060005AZ

COMPOSITION:
Each vial contains
Cefoperazone Sodium / Sulbactam Sodium equivalent to Cefoperazone 500mg and Sulbactam 500mg.

PRESENTATION:
White or off-white powder free from visible contamination. Clear to pale yellow solution after reconstitution.

INDICATIONS:
Monotherapy:
Sulbactam / cefoperazone is indicated for the treatment of the following infections when caused by susceptible organism:
Respiratory tract infections (upper and lower). Urinary tract infections (upper and lower). Peritonitis. Cholecystitis. Cholangitis and other intra-abdominal infections. Septicemia. Meningitis. Skin and soft tissue infections. Bone and joint infections. Pelvic Inflammatory Disease. Endometritis. Gonorrhea and other infections of the genital tract.

Combination Therapy:
Because of the broad spectrum of the activity of sulbactam/cefoperazone, most infections can be treated adequately with this antibiotic alone. However, sulbactam / cefoperazone may be used concomitantly with other antibiotics if such combinations are indicated. If an aminoglycoside is used (see Interaction With Other Medicaments), renal function should be monitored during the course of therapy (see Recommended Dose Use in Renal Function).

PHARMACODYNAMICS:
The antibacterial component of sulbactam / cefoperazone is cefoperazone, a third generation cephalosporin, which acts against sensitive organisms during the stage of active multiplication by inhibiting biosynthesis of cell wall mucopeptide. Sulbactam does not possess any useful antibacterial activity, except against Neisseriaceae and Acinetobacter. However, biochemical studies with cell-free bacterial systems have shown it to be an irreversible inhibitor of most important b - lactamases produced by b-lactam antibiotic - resistant organisms.
The potential for sulbactam preventing the destruction of penicillins and cephalosporins by resistant organisms was confirmed in whole-organism studies using resistant strains in which sulbactam exhibited marked synergy with penicillins and cephalosporins. As sulbactam also binds with some penicillin-binding proteins, sensitive strains are also often rendered more susceptible to sulbactam / cefoperazone than to cefoperazone alone.
The combination of sulbactam and cefoperazone is active against all organisms sensitive to cefoperazone. In addition, it demonstrates synergistic activity (up to 4-fold reduction in minimum inhibitory concentrations for the combination versus those for each component) in a variety of organisms, most markedly the following: *Haemophilus influenzae, Bacteroides* sp, *Staphylococcus* sp, *Acinetobacter calcoaceticus, Enterobacter aerogenes, Escherichia coli, Proteus mirabilis, Klebsiella pneumoniae, Morganella morganii, Citrobacter freundii, Enterobacter cloacae, Citrobacter diversus.*
Sulbactam / cefoperazone is active *in vitro* against a wide variety of clinically significant organisms: Gram-Positive Organisms: *Staphylococcus aureus* (penicillinase - and non penicillinase - producing strains), *Staphylococcus epidermidis, Streptococcus pneumoniae* (formerly *Diplococcus pneumoniae*), *Streptococcus pyogenes* (Group A b-hemolytic streptococci), *Streptococcus agalactiae* (Group B b-hemolytic streptococci), most other strains of b-hemolytic streptococci, many strains of *Streptococcus faecalis* (enterococcus).
Gram-Negative Organisms: *Escherichia coli, Klebsiella* sp, *Enterobacter* sp and *Citrobacter* sp, *Haemophilus influenzae, Proteus mirabilis, Proteus vulgaris, Morganella morganii* (formerly Proteus morganii), *Providencia rettgeri* (formerly *Proteus rettgeri*), *Providencia* sp, *Serratia* sp (including *S. marcescens*), *Salmonella* sp, *Shigella* sp, *Pseudomonas aeruginosa* and some other *Pseudomonas* sp, *Acinetobacter calcoaceticus, Neisseria gonorrhoeae, Neisseria meningitidis, Bordetella pertussis* and *Yersinia enterocolitica*.
Anaerobic Organisms: Gram-negative bacilli (including *Bacteroides fragilis*, other *Bacteroides* sp, and *Fusobacterium* sp).
Gram-positive and gram-negative cocci (including *Peptococcus, Peptostreptococcus* and *Veillonella* sp).
Gram-positive bacilli (including Clostridium, Eubacterium and *Lactobacillus* sp).

Susceptibility Testing: The following susceptibility ranges have been established for sulbactam / cefoperazone : (See Table 1).

	Minimal Inhibitory Concentration (MIC), (mcg/mL - expressed as cefoperazone concentration)
Susceptible Intermediate Resistant	≤16 17-63 ≥ 64
	Susceptibility Disc Zone Size, mm (Kirby - Bauer)
Susceptible Intermediate Resistant	≥ 21 16-20 ≤ 15

For MIC determinations, serial dilutions of sulbactam / cefoperazone may be used with a broth or agar dilution method. Use of a susceptibility test disc containing 30mcg of sulbactam and 75mcg of cefoperazone is recommended. A report from the laboratory of "susceptible" indicates that the infecting organism is likely to respond to sulbactam / cefoperazone therapy, and a report of "resistant" indicates that the organism is not likely to respond. A report of "intermediate" suggests that the organism would be susceptible to sulbactam / cefoperazone if a higher dosage is used or if the infection is confined to tissues or fluids where high antibiotic levels are attained.

The following quality control limits are recommended for 30mcg / 75mcg sulbactam / cefoperazone susceptibility discs. (See Table 2).

Control Strain	Zone Size (mm)
Acinetobacter sp, ATCC 43498	26-32
Pseudomonas aeruginosa, ATCC 27853	22-28
Escherichia coli, ATCC 25922	27-33
Staphylococcus aureus, ATCC 25923	23-30

PHARMACOKINETICS:
Approximately 84% of the sulbactam dose and 25% of the cefoperazone dose administered with sulbactam / cefoperazone is excreted by the kidney. Most of the remaining dose of cefoperazone is excreted in the bile. After sulbactam/cefoperazone administration, the mean half-life for sulbactam is about 1 hr while that for cefoperazone is 1.7 hrs. Serum concentrations have been shown to be proportional to the dose administered. These values are consistent with previously published values for the agents when given alone. Mean peak sulbactam and cefoperazone concentrations after the administration of 2g of sulbactam / cefoperazone (1g sulbactam, 1g cefoperazone) IV over 5 min were 130.2 and 236.8mcg/mL, respectively. This reflects the larger volume of distribution for sulbactam (Vd=18-27.6L) compared to cefoperazone (Vd=10.2-11.3 L).
After IM administration of 1.5g sulbactam / cefoperazone (0.5g sulbactam, 1g cefoperazone) peak serum concentrations of sulbactam and cefoperazone are seen from 15 min to 2 hrs after administration. Mean peak serum concentrations were 19 and 64.2 mcg/mL for sulbactam and cefoperazone, respectively. Both sulbactam and cefoperazone distribute well into a variety of tissues and fluid including bile, gall bladder, skin, appendix, fallopian tubes, ovary, uterus, and others. There is no evidence of any pharmacokinetics drug interaction between sulbactam and cefoperazone when administered together in the form of sulbactam / cefoperazone. After multiple dosing, no significant changes in the pharmacokinetics of either component of sulbactam / cefoperazone have been reported and no accumulation has been observed when administered every 8-12 hrs.

Hepatic Dysfunction: See Warning and Precautions.

Renal Dysfunction: In patients with different degrees of renal function administered sulbactam / cefoperazone, the total body clearance of sulbactam was highly correlated with estimated creatinine clearance. Patients who are functionally anephric, showed a significantly longer half-life of sulbactam (mean 6.9 and 9.7 hrs in separate studies). Hemodialysis significantly altered the half-life, total body clearance and volume of distribution of sulbactam. No significant differences have been observed in the pharmacokinetics of cefoperazone in renal failure patients.

Elderly: The pharmacokinetics of sulbactam / cefoperazone have been studied in elderly individuals with renal insufficiency and compromised hepatic function. Both sulbactam and cefoperazone exhibited longer half-life, lower clearance and larger volumes of distribution when compared to data from normal volunteers. The pharmacokinetics of sulbactam correlated well with the degree of renal dysfunction while for cefoperazone there was a good correlation with the degree of hepatic dysfunction.

Children: Studies conducted in pediatrics have shown no significant changes in the pharmacokinetics of the components of this product compared to adult values. The mean half-life in children has ranged from 0.91-1.42 hrs for sulbactam and from 1.44-1.88 hrs for cefoperazone.

DOSAGE AND ADMINISTRATION:
Use in Adults:
Daily dosage recommendations for sulbactam / cefoperazone in adults are as follows:

Ratio	Cefoperazone / Sulbactam (g)	Sulbactam Activity (g)	Cefoperazone Activity (g)
1:1	2.0 - 4.0	1.0 - 2.0	1.0 - 2.0

Doses should be administered every 12 hours in equally divided doses. In severe or refractory infections the daily dosage of cefoperazone / sulbactam may be increased up to 8g (i.e., 4g cefoperazone activity). Patients maybe require additional cefoperazone administered separately. Doses should be administered every 12 hours in equally divided doses. The recommended maximum daily dosage of sulbactam is 4g.

Use in Hepatic Dysfunction:
See Warning and Precaution

Use in Renal Dysfunction:
Dosage regimens of cefoperazone / sulbactam should be adjusted in patients with marked decrease in renal function (creatinine clearance of less than 30ml/min) to compensate for the reduced clearance of sulbactam. Patients with creatinine clearances between 15 and 30ml/min should received a maximum of 1g of sulbactam administered every 12 hours (maximum daily dosage of 2g sulbactam), while patients with creatinine clearance of less than 15ml/min should receive a maximum of 500mg of sulbactam every 12 hours (maximum daily dosage of 1g sulbactam). In severe infections it may be necessary to administer additional cefoperazone. The pharmacokinetic profile of sulbactam is significantly altered by hemodialysis. The serum half-life of cefoperazone is reduced slightly during hemodialysis. Thus, dosing should be scheduled to follow a dialysis period.

Use in Elderly:
See Pharmacokinetic Properties

Use in Children:
Daily dosage recommendations for cefoperazone / sulbactam in children are as follows:

Ratio	Cefoperazone / Sulbactam (mg/kg/day)	Sulbactam Activity (mg/kg/day)	Cefoperazone Activity (mg/kg/day)
1:1	40 - 80	20 - 40	20 - 40

Doses should be administered every 6 to 12 hours in equally divided doses. In serious or refractory infections, these dosages may be increased up to 160mg/kg/day. Doses should be administered in two to four equally divided doses (see Warning and Precautions for *Use in Infancy* and *Use in Pediatrics*).

Use in Neonates:
For neonates in the first week of life, the drug should be given every 12 hours. The maximum daily dosage of sulbactam in pediatrics should not exceed 80mg/kg/day. For doses of sulbactam/cefoperazone requiring more than 80mg/kg/day cefoperazone activity, additional cefoperazone should be administered.

Intravenous Administration:
For intermittent infusion, each vial of sulbactam / cefoperazone should be reconstituted with the appropriate amount (see Instruction for Use / Handling Reconstitution) of 5% Dextrose in Water, 0.9% Sodium Chloride Injection or Sterile Water for Injection and then diluted to 20ml with the same solution followed by administration over 15-60 min. Lactated Ringer's Solution is a suitable vehicle for intravenous infusion, however, not for initial reconstitution (see Interaction With Other Medicaments). For intravenous injection, each vial should be reconstituted as described previously and administered over a minimum of 3 minutes.

Intramuscular Administration:
Lidocaine HCl 2% is a suitable vehicle for IM administration, however, not for initial reconstitution (see Interaction With Other Medicaments *Lidocaine* and *Instruction of use / handling Lidocaine*).

Instruction of use / handling:
Reconstitution

Total Dosage (g)	Equiv. Dosage of Sulb. + cefoperazone (g)	Volume of diluent	Maximum Final Conc. (mg/ml)
1.0	0.5 + 0.5	3.4	125 + 125

Sulbactam / cefoperazone has been shown to be compatible with water for injection, 5% dextrose, normal saline, 5% dextrose in 0.225% saline, 5% dextrose in normal saline, Lactated Ringer's solution and lidocaine 2% injection, which stable for 24 hours.

Lactated Ringer's Solution: Sterile water for injection should be used for reconstitution (see Interactions with Other Medicament: Incompatibilities). A 2-step dilution is required using sterile water for injection, further diluted with Lactated Ringer's solution to a sulbactam concentration of 5mg/ml (use 2ml initial dilution in 50ml or 4ml initial dilution in 100ml Lactated Ringer's solution).

Lidocaine: Sterile water for injection should be used for reconstitution (see Interactions with Other Medicament: Incompatibilities). A 2-step dilution is required using sterile water for injection, further diluted with 2% lidocaine to yield solutions containing up to 125mg cefoperazone and 125mg sulbactam/ml in approximately a 0.5% lidocaine HCl solution.

ROUTE OF ADMINISTRATIONS:
For intramuscular and intravenous administration only.

CONTRAINDICATIONS:
Sulbactam / cefoperazone is contraindicated in patients with known allergy to penicillins, sulbactam, cefoperazone or any of the cephalosporins.

SIDE EFFECTS:
Sulbactam / cefoperazone is generally well tolerated. The majority of adverse events are of mild or moderate severity and are tolerated with continued treatment.

Gastrointestinal: As with other antibiotics, the most frequent side effects observed with sulbactam / cefoperazone have been gastrointestinal. Diarrhea / loose stools have been reported most frequently followed by nausea and vomiting.

Dermatologic Reactions: As with all penicillins and cephalosporins, hypersensitivity manifested by maculopapular rash, urticaria has been reported. These reactions are more likely to occur in patients with a history of allergies, particularly to penicillin.

Hematology: Slight decreases in neutrophils have been reported. As with other β-lactam antibiotics, reversible neutropenia may occur with prolonged administration. Some individuals have developed a positive direct Coombs' test during treatment. Decreased hemoglobin or hematocrit have been reported, which is consistent with published literature on cephalosporins. Transient eosinophilia and thrombocytopenia have occurred and hypoprothrombinemia has been reported.

Miscellaneous: Headache, fever, injection pain and chills occurred in of patients.

Laboratory Abnormalities: Transient elevations of liver function tests, SGOT, SGPT, alkaline phosphatase and bilirubin levels have been noted.

Local Reactions: Sulbactam / cefoperazone is well tolerated following IM administration. Occasionally, transient pain may follow administration by this route. As with other cephalosporins and penicillins, when sulbactam/cefoperazone is administered by an IV catheter some patients may develop phlebitis at the infusion site.

In post-marketing experience, the following additional undesirable effects have been reported:
General: Anaphylactoid reaction (including shock).
Cardiovascular: Hypotension.
Gastrointestinal: Pseudomembranous colitis.
Skin / Appendages: Pruritus, Stevens-Johnson syndrome.
Hematopoietic: Leucopenia.
Urinary: Hematuria.

SYMPTOMS AND TREATMENTS FOR OVERDOSAGE:
Limited information is available on the acute toxicity of cefoperazone sodium and sulbactam sodium in humans. Overdosage of the drug would be expected to produce manifestations that are principally extensions of the adverse reactions reported with the drug. The fact that high CSF concentrations of β-lactam antibiotics may cause neurologic effects, including seizures, should be considered. Because cefoperazone and sulbactam are both removed from the circulation by hemodialysis, these procedures may enhance the elimination of the drug from the body if overdosage occurs in patients with impaired renal function.

WARNING AND PRECAUTION:
Hypersensitivity:
Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactams agents. If allergic reaction occurs, the product must be discontinued immediately and appropriate alternative therapy instituted. Serious anaphylactic reactions require immediate emergency treatment with epinephrine. Oxygen, intravenous steroid and airway management, including intubation, should be administered as indicated.

Use In Hepatic Dysfunction:
Cefoperazone is extensively excreted in bile. The serum half-life of cefoperazone is usually prolonged and urinary excretion of the drug increased in patients with hepatic diseases and / or biliary obstruction. Even with severe hepatic dysfunction, therapeutic concentrations of cefoperazone are obtained in bile and only a 2- to 4- fold increase in half-life is seen. Dose modification may be necessary in case of severe biliary obstruction, severe hepatic disease or in cases of renal dysfunction coexistent with either of those conditions. In patients with hepatic dysfunction and concomitant renal impairment, cefoperazone serum concentrations should be monitored and dosage adjusted as necessary. In these cases, dosage should not exceed 2g/day of cefoperazone without close monitoring of serum concentrations.

General:
As with other antibiotics, vitamin K deficiency has occurred in a few patients treated with cefoperazone. The mechanism is most probably related to the suppression of gut flora which normally synthesize this vitamin. Those at risk include patients with poor diet, malabsorption states (e.g., cystic fibrosis) and patients on prolonged intravenous alimentation regimens. Prothrombin time should be monitored in these patients receiving anticoagulant therapy and exogenous vitamin K administered as indicated. As with other antibiotics, overgrowth of nonsusceptible organisms may occur during prolonged use of sulbactam / cefoperazone. Patients should be observed carefully during treatment. As with any potent systemic agent, it is advisable to check periodically for organ system dysfunction during extended therapy; this includes renal, hepatic and hematopoietic systems. This is particularly important in neonates, especially when premature and other infants.

Use In Infancy:
Sulbactam / cefoperazone has been effectively used in infants. It has not been extensively studied in premature infants or neonates. Therefore, in treating premature infants and neonates potential benefits and possible risks involved should be considered before instituting therapy. Cefoperazone does not displace bilirubin from plasma protein binding sites.

Use During Pregnancy:
Sulbactam and cefoperazone cross the placental barrier. There are, however, no adequate and well-controlled studies in pregnant women. Because animal reproduction studies are not always predictive of human response, this drug should be used during pregnancy only if clearly needed.

Use In Nursing Mother:
Only small quantities of sulbactam and cefoperazone are excreted in human milk. Although both drugs pass poorly into breast milk of nursing mothers, caution should be exercised when sulbactam / cefoperazone is administered to a nursing mother.

INTERACTION WITH OTHER MEDICAMENTS:
Alcohol:
A reaction characterized by flushing, sweating, headache and tachycardia has been reported when alcohol was ingested during and as late as the fifth day after cefoperazone administration. A similar reaction has been reported with certain other cephalosporins and patients should be cautioned concerning ingestion of alcoholic beverages in conjunction with administration of sulbactam / cefoperazone. For patients requiring artificial feeding orally or parenterally, solutions containing ethanol should be avoided.

Drug Laboratory Test Interactions:
A false positive reaction for glucose in the urine may occur with Benedict's or Fehling's solution.

Incompatibilities:
Aminoglycosides:
Solutions of sulbactam / cefoperazone and aminoglycosides should not be directly mixed, since there is a physical incompatibility between them. If combination therapy with sulbactam / cefoperazone and an aminoglycoside is contemplated (see Indications *Combination Therapy*) this can be accomplished by sequential intermittent intravenous infusion provided that separate secondary intravenous tubing is used, and that the primary intravenous tubing is adequately irrigated with approved diluents between doses. It is also suggested that doses of sulbactam / cefoperazone be administered throughout the day at times as far removed from administration of the aminoglycoside as possible.

Lactated Ringer's Solution:
Initial reconstitution with Lactated Ringer's Solution should be avoided since these mixtures have been shown to be incompatible. However, a 2-step dilution process involving initial reconstitution in water for injection will result in a compatible mixture when further diluted with Lactated Ringer's Solution (see Dosage & Administration: *Reconstitution*).

Lidocaine:
Initial reconstitution with 2% lidocaine HCl solution should be avoided since these mixture has been shown to be incompatible. However, a 2-step dilution process involving initial reconstitution in water for injection will result in a compatible mixture when further diluted with 2% lidocaine HCl (see Dosage & Administration: *Instruction for use / handling*).

STORAGE:
Store below 30°C. Protect from light.
Reconstituted solution stable for 1 day at storage below 25°C.

KEEP OUT OF REACH OF CHILDREN
JAUHI DARI KANAK-KANAK

PACK QUANTITIES:
Single vial per box and 10 x 1g per box.

Further information can be obtained from pharmacist, physician or the manufacturer.

Product Registration Holder & Manufactured By:
KOTRA PHARMA
Kotra Pharma (M) Sdn. Bhd.
No.1, 2 & 3, Jalan TTC 12, (90082-V)
Cheng Industrial Estate,
75250 Melaka, Malaysia.

Date of revision : 29/12/21